

Configuration management

1 Formal Description

Identifier	Name	Owner	User ID
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Process Designers

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1.1 Scope for Application of this process

 **To describe here:** For which projects / project types or organizational units and at which times or phases this process applies

1.1.1 Scope of Domain and Configuration Library

Scope :□This process is applied to all domains to participated project.

The documents which is generated by each domain are consolidated by a project.

Other configuration items are managed according to configuration plan which is defined by configuration library, because those configuration items has different library for each domain.

The configuration management details for each domain are detailed in the domain-specific configuration management plan.(e.g. Software Configuration Management Plan)

スコープ：本プロセスは、プロジェクトに関わる全てのドメインに適用されます。

各ドメインによって作成されるドキュメントはプロジェクトで一元管理されます。

その他の構成品目はドメイン毎に構成管理ライブラリが異なるため、構成管理ライブラリ別に定義される構成管理計画書に従って管理されます。

ドメイン毎の構成管理の詳細は、ドメイン別の構成管理計画書に記載されます。

Table

Domain	Configuration Library						
	Redmine	SVN(Doc)	SVN(Src)	Git(Src)	EA	3DEX	DS2
Software	X	X	X	X	X		
Hardware	X	X					(x)
Mechanical	X	X				(x)	
Quality Assurance	X	X					
Reliability	X	X	X				
Functional Safety	X	X					
SQA	X	X					

Industrialisation	X	X	X				
Production	X	X	X				
Purchasing	X						
Logistic	X	X					
Sales/Marketing	X						
Service	X						
ClaimAnalysis	X						
Project Infrastructure	X						

1.1.2 Life Cycle

Configuration Management shall include following time frames or phase of product life cycle.

- Start: RFI/RFQ Phase
- End: End of Field Service Phase

Configuration Managementは以下の製品ライフサイクルを対象とする。

- Start: RFI/RFQ フェーズ
- End: フィールドサービスフェーズ

1.2. Tailoring Guideline

1.3 Relations

 **To describe here:** The relation with other processes / standards / regulations or policies

- Relations
 - Other processes
 - PMS
 - Project Management Sub Process
 - Quality Management Sub Process
 - SyIVE
 - Standards
 - A-SPICE 3.1
 - SUP.8 Configuration Management
 - IEEE Standards
 - IEEE828-2012 Standard for Configuration Management in Systems and Software Engineering
 - Unified QS
 - [SUP.8 Configuration Management](#)

2 Strategy

2.1. Purpose

Purpose of the Configuration Management Process is as follows.

- Provide a configuration management process and infrastructure which is easy to use for users.
- Ensure the management of the data needed to re-build of the product.

Configuration Management process は、下記を目的とする

- ユーザーに負担がなく、利用しやすい構成管理プロセス・インフラを提供する。
- 製品の再構築に必要なデータの管理を保証する。

2.2. Objective/Goal

Objective/Goal of the CM process is as bellow.

- All projects shall meet the level3 requirements of A-SPICE.
- Development environment shall be recovered promptly.
- Several model/Variant can be developed smoothly.
- Status of configuration items shall be reported to management regularly.
- Baseline registration shall be finished in 2 weeks from Quality Gate review.
- Storage of management data shall be ensured.
- It can be recovered promptly when configuration management environment destroyed by Disaster.
- If project stakeholder needs to refer configuration items, it can be easy for them to access.
- Prevent the unauthorized access to configuration library.

CM Processは、下記をObjective/Goalとする

- すべてのプロジェクトでA-SPICE Level3の要求を満たす。
- 開発環境の復元が速やかにできる。
- 複数モデル/Variantの開発がスムーズに行える。
- 構成品目のステータスがマネジメントに定期的に報告される。
- Q-Gateレビュー開催からベースライン登録完了までを2Week以内に実施する。
- 管理データの保管が保証される。
- 災害発生による構成管理環境の破壊発生時でも、速やかに復旧させることができる。
- プロジェクト関係者が参照したい構成品目に容易にアクセスできる。
- 構成ライブラリへの不正アクセスを防ぐ。

2.3. Strategy

CM has set following strategy to reach Objective/Goals.

- Generate the CM Plan according to Template/Guideline.
- Manage configuration items using DRP in all domains.
- Manage a Baseline.
- Control a Rollback
- Control a Branch and Merge.
- Report status of configuration item.
- Perform Baseline audit/Configuration audit.
- Ensure a storage of defined scope.

CMは目標/ゴールを達成する為に以下の戦略を設定する。

- Template/Guidelineに従ってCM Planを作成する。
- 全てのドメインでDRPを用いた構成品目の管理を行う。
- ベースライン管理を行う。
- Rollback制御を行う。
- ブランチマージを制御する。
- 構成品目のステータスを報告する。
- 構成管理に対する監査を実施する。
- ベースライン監査/構成監査を実施する。

- Perform a recover training.
- Provide a configuration management system.
- Manage an access authority.
- Monitor an access log.
- 定められたScope内の保管を保証する。
- 復旧訓練を実施する。
- 構成管理システムを提供する。
- アクセス権の管理を行う。
- アクセスログの監視を行う。

3 Stakeholders

3.1 Skills, Capabilities, Responsibility & Authority

 **To describe here:** which main roles for this process do exist and what is their general responsibility with regard to the overall process. For each role the generic Skills and Capabilities shall be defined. The definition of the roles is part of the EA/Activity Model

Role	Skills	Capabilities	Responsibility	Authority
Configuration Manager	Education program for Configuration Management	1.)Expert knowledge of Configuration Management	- Understand Configuration requirements - Confirm unclear points to Configuration - Create configuration environment - Establish baselines - Report Configuration Status to stakeholders	Decide configuration management rules and to make member follow it.
Configuration Management assistant	Education program for Configuration Management tool	1.)knowledge of Configuration Management library	- Review work products - Judgement of review result	Operation of configuration management tool
Domain Configuration Manager	Education program for own domain Configuration Management	1.)Expert knowledge of Configuration Management of own domain	□Create domain CM Plan □Create Guideline □Judge result of physical Configuration audit	Decide domain configuration management rules and to make member follow it.
Domain Configuration Management assistant	Education program for own domain Configuration Management tool	1.)knowledge of Configuration Management library	- Review work products of SW - Judgement of review result of SW	Operation of configuration management tool

Role	Skills	Capabilities	Responsibility	Authority
Site Configuration Manager	System Infrastructure knowledge	Expert knowledge of Infrastructure	- Develop Repository Server - Backup Data	- Construction of server about configuration management - Operation of backup data management system
Project Infrastructure development	System Infrastructure knowledge	Expert knowledge of Project Infrastructures	- Setup project infrastructure	Set up the defect management tools (Synchronization with Customer / supplier)
Project Leader	1.)PMP 2.)Common language with customer 3.)Management of the change requests	1.)Negotiation and coordination with the customer 2.)Cordination with domain leaders 3.)Cordination with internal stakeholders 4.) Negotiation about change contents and shcedule with CR issuer	1.)all deliverables of project 2.)Approve the estimation for the change contents 3.)Responsibility the quality of product/project	Authority to provide CM Environment Authority to assign CM/CM Staff Authority to approve CM Plan
Quality Assurance	1.)Education program for Quality Assurance 2.)Common language with customer 3.)A-SPICE	1.) Understanding quality requirement of Project 2.) Ability to judge an influence degree of the quality	- Escalation to general manager	1.) Authority to assign Domain QA/Domain QA Staff 2.) Authority to approve Domain QA Plan 3.) Authority to judge release by the quality point of view

3.2 Stakeholder involvement

 **To describe here:** Define how the main stakeholders of this process are involved in a generic manner. This involvement typically needs to be tailored while setting up a project. The Stakeholder involvement is defined in the EA/Activity Model.

Activity	Method /Type	Frequency	CM	CM Staff	SCM	SCM Staff	PL	Quality Assurance	Site CM
Do Configuration Item Audit			A	R	A	R	I	I	
Responsible(責任), Approving Authority(承認), Support(支援), Informed(通知) and Consulted(相談)									

 **Note:** For example in the case of small project , it is permitted that Domain Leader makes Domain Configuration Management Plan.

4 Input Work Products

- Project SOW
- プロジェクト SoW

5 Output Work products

- Configuration Management Plan□DRP
- Configuration Management Plan□DRP Review record
- Configuration Status Report
- Project Deliverables
- Baseline Result
- Configuration Audit Report
- 構成管理計画書□DRP
- 構成管理計画書□DRPレビュー記録
- 構成管理ステータスレポート
- プロジェクト成果物
- ベースライン結果
- 構成監査レポート

6 Process Practices

 **To describe here:** The process activities are described here together with the respective BPMN diagrams maintained in the EA/Activity Model

6.1 Life Cycle relation



開発フェーズ

6.2 Sequence & interaction with other processes



他Processとのシグナル

6.3 Entry & Exit conditions (triggers)

- **Start triggers**
 - Request domain specific Plan
- **end triggers**
 - Completion of configuration management plan and preparation
 - Distribute configuration management plan
 - Start a quality audit
 - Complete the configuration management release cycle.
- **Start Conditions**
 - Release a Project Charter from Project Leader.
- **開始トリガー**
 - ドメイン計画の要求
- **終了トリガー**
 - 構成管理計画および準備の完了
 - 構成管理計画書の展開
 - 品質監査の開始
 - 構成管理のリリースサイクルの完了
- **Start Conditions**
 - Project LeaderからのProject Charterのリリース
- **Exit Conditions**
 - プロジェクトで計画された最終

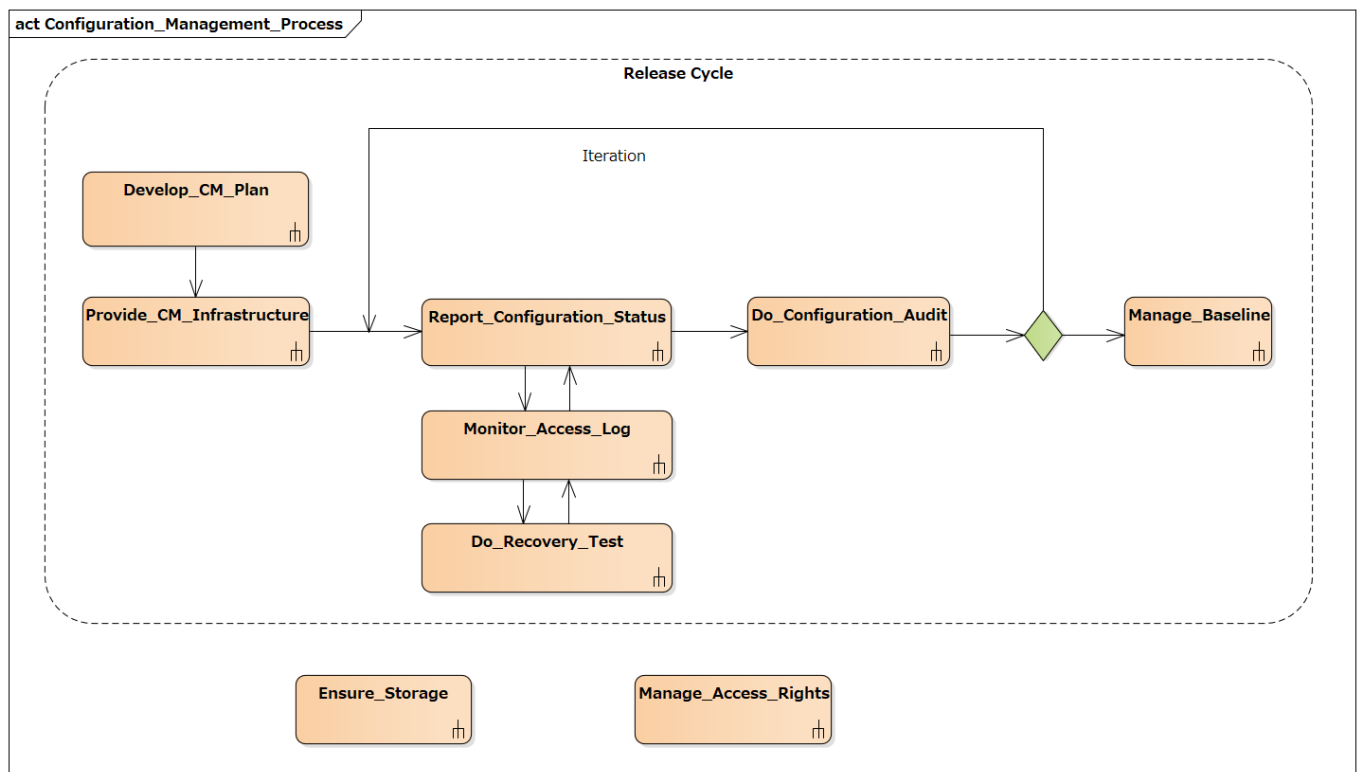
- **Exit Conditions**

- When Baseline of last Milestone Event which is planned in the project has finished.
- When the project cancelled.

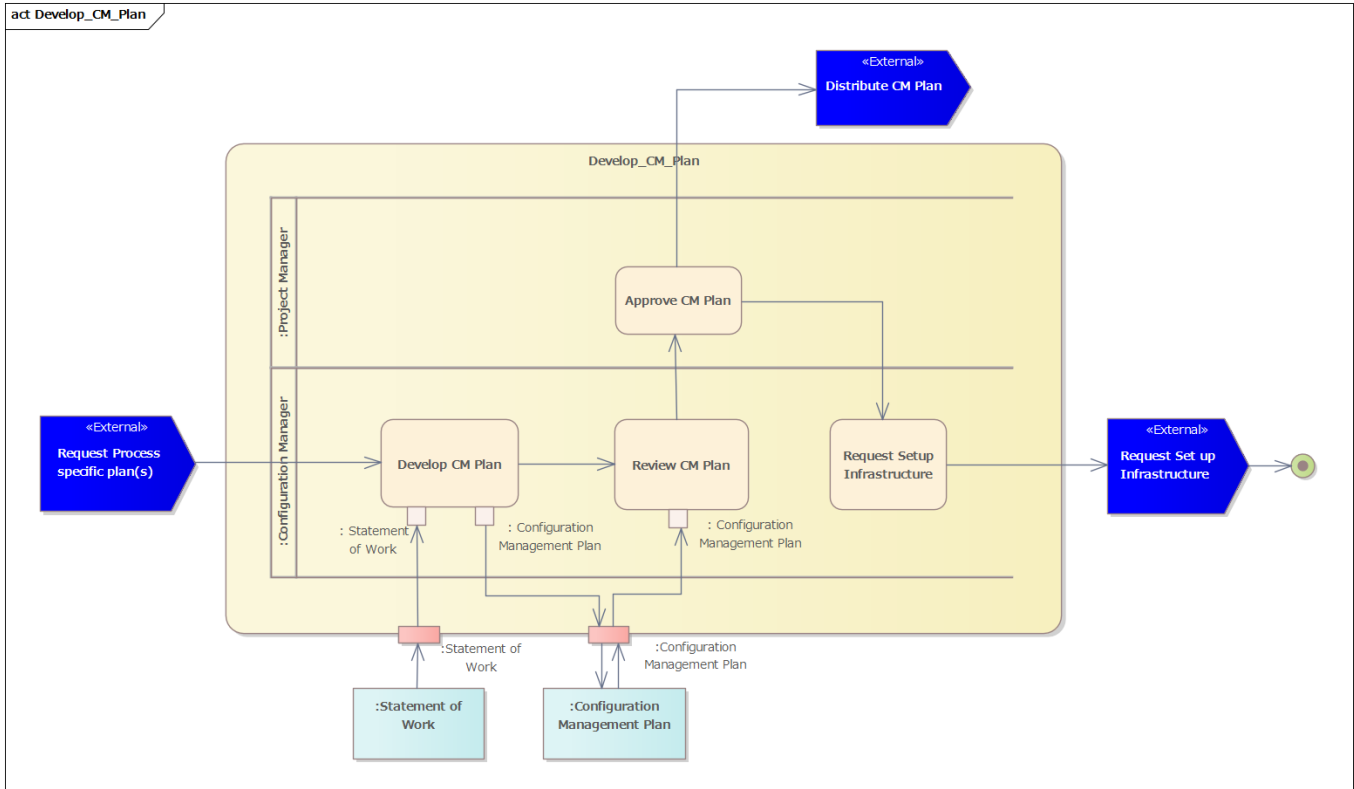
のMilestone Eventのベースライン化が完了したとき。
◦ プロジェクトが中止されたとき。

6.4 Process Flow

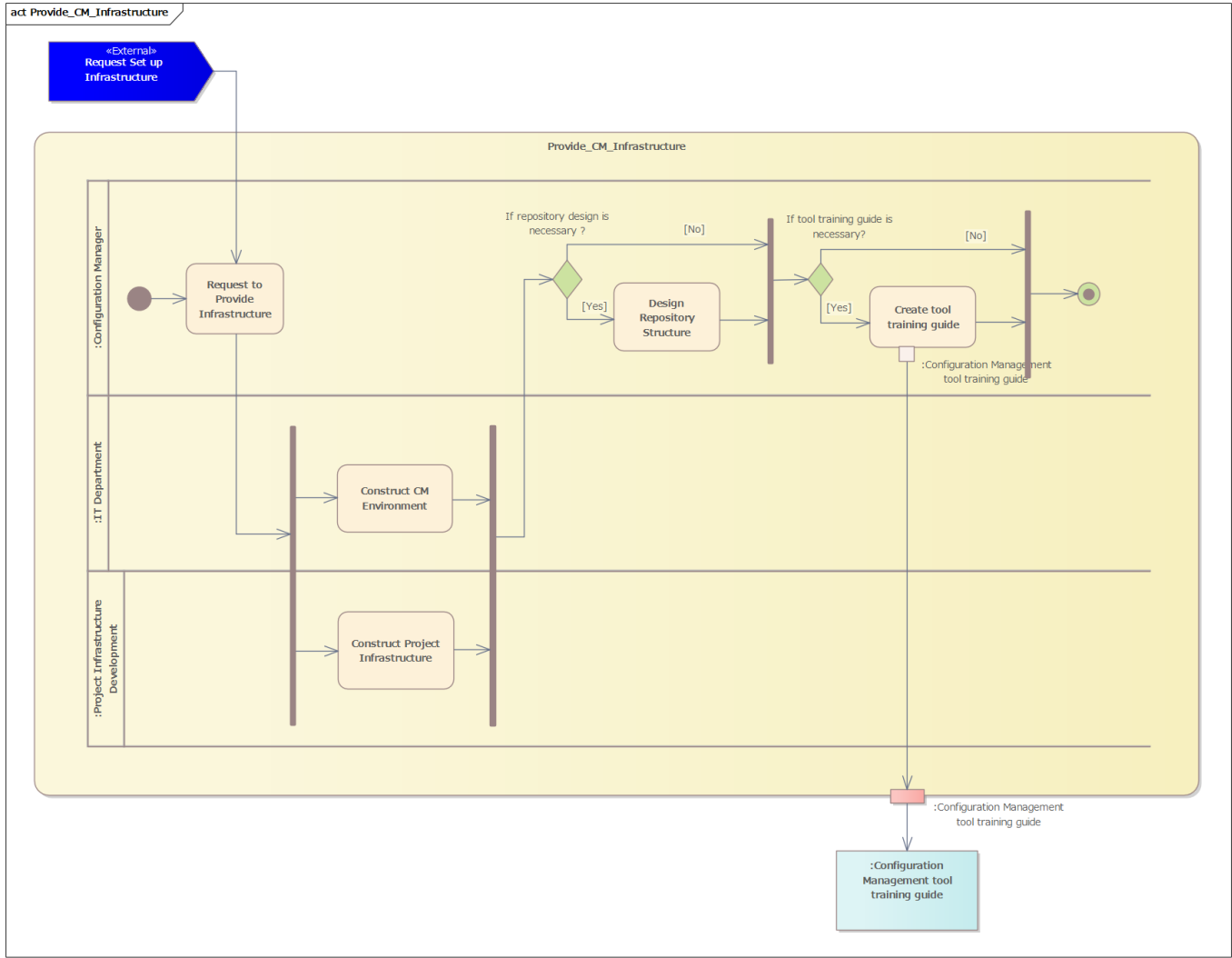
6.4.1 Configuration Management Process Overview



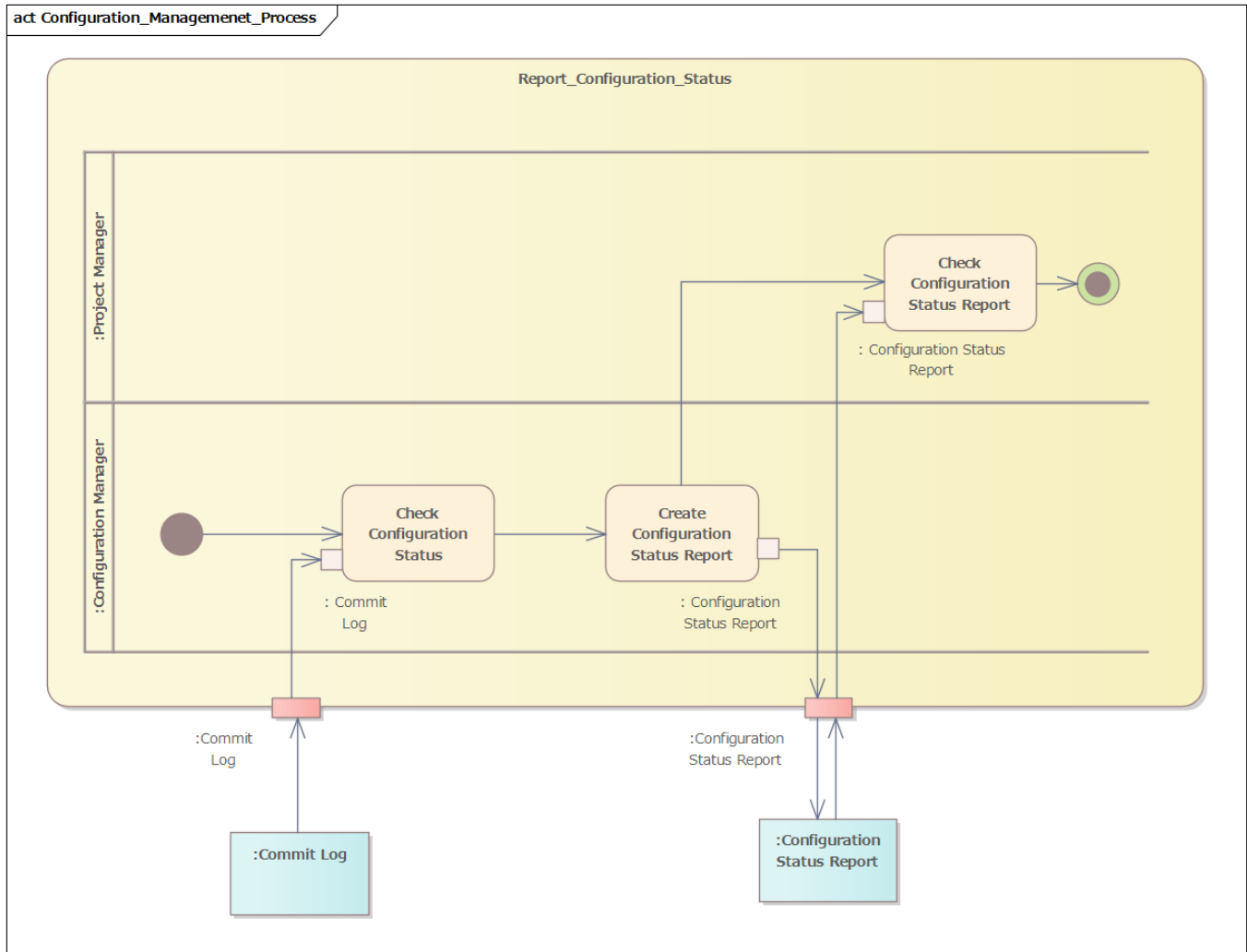
6.4.2 Develop Configuration Management Plan



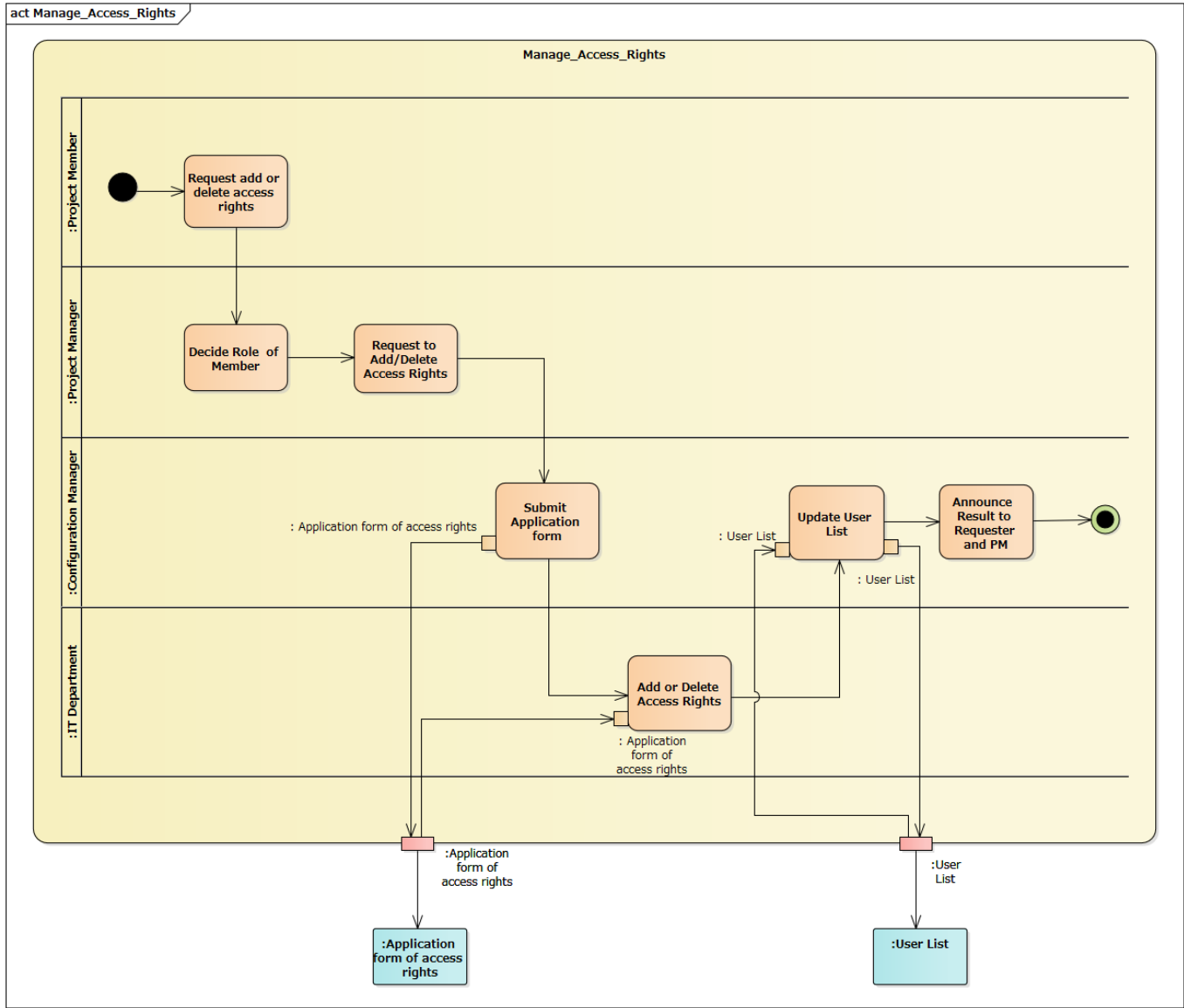
6.4.3 Provide CM Infrastrucure



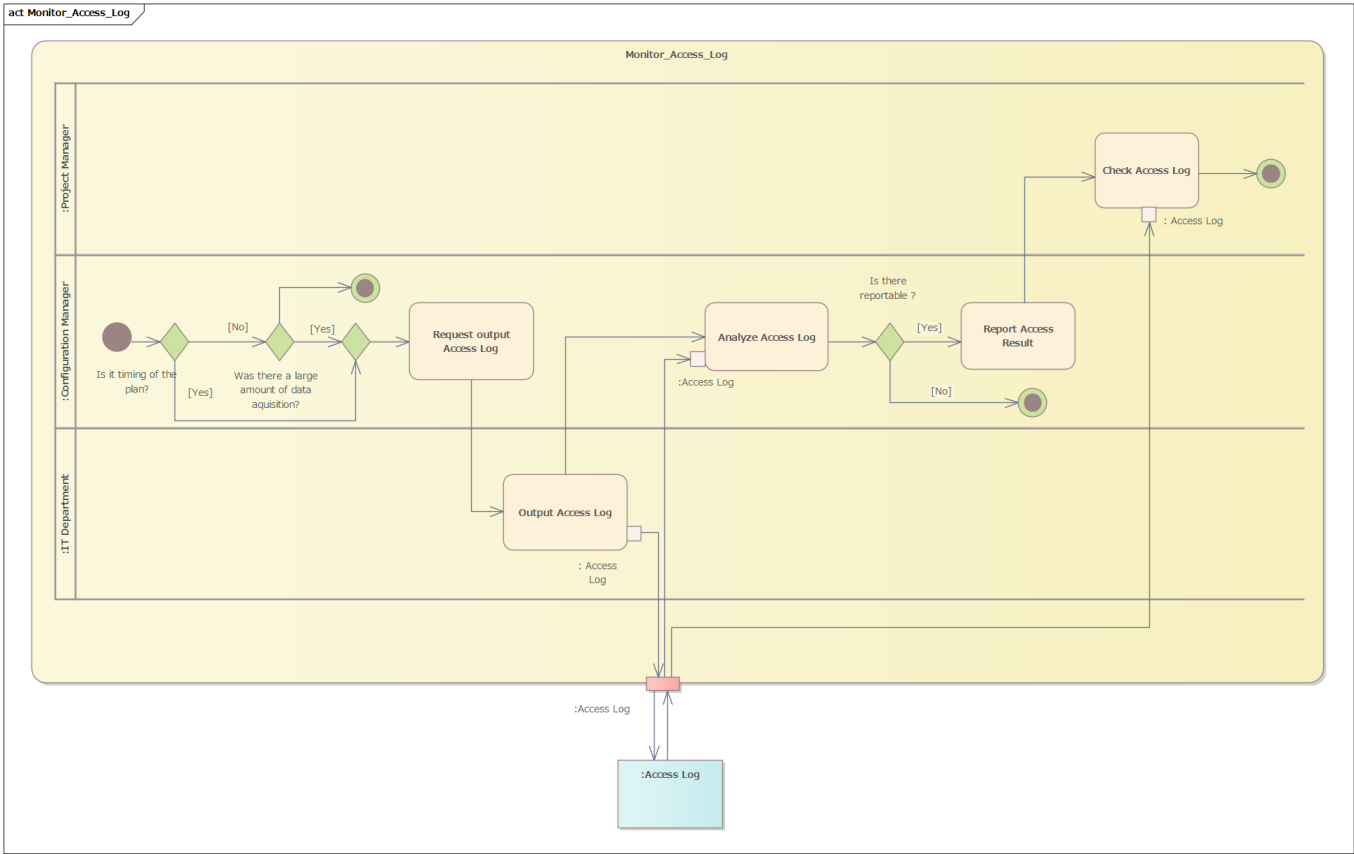
6.4.4 Report Configuration Status



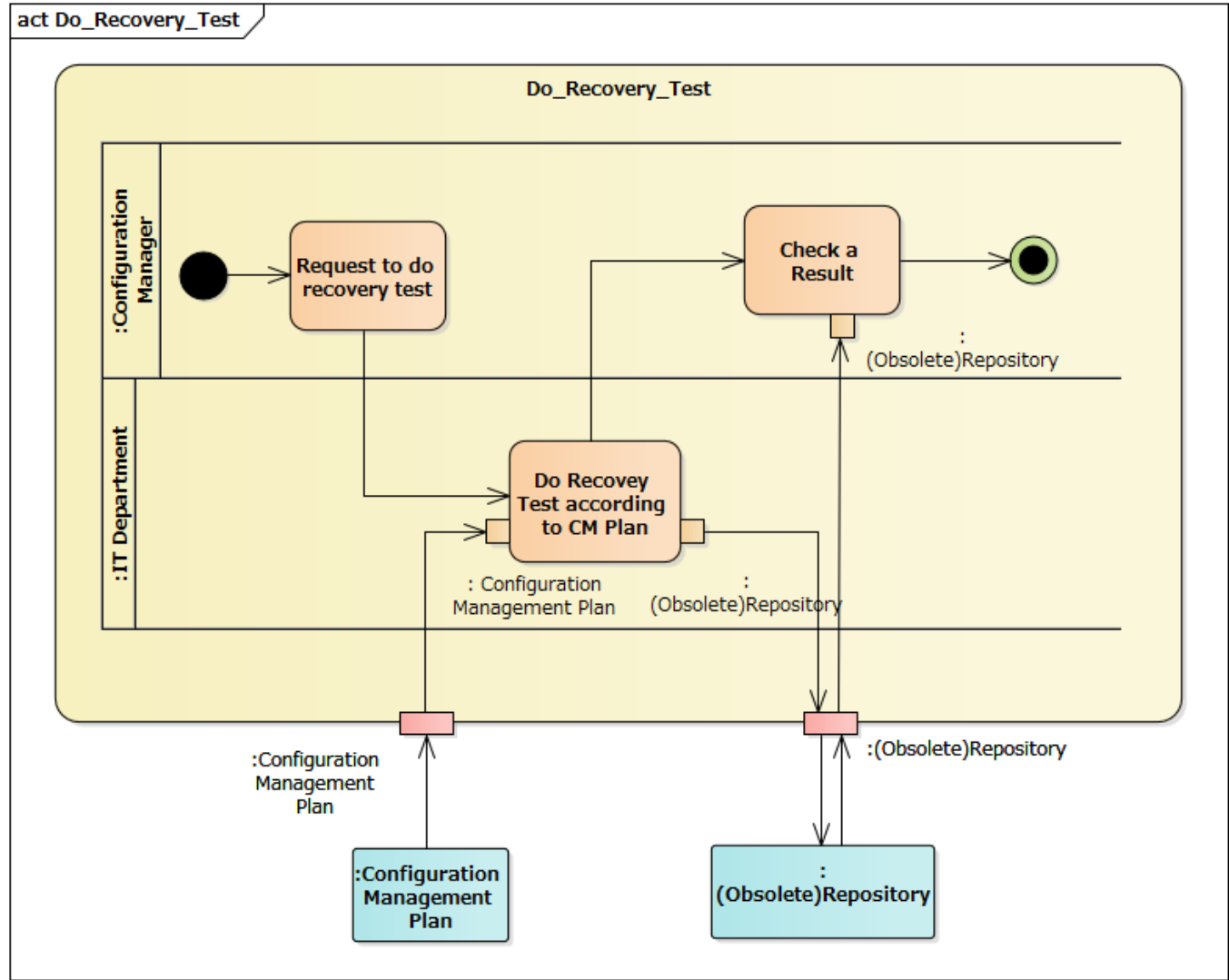
6.4.5 Manage Access Rights



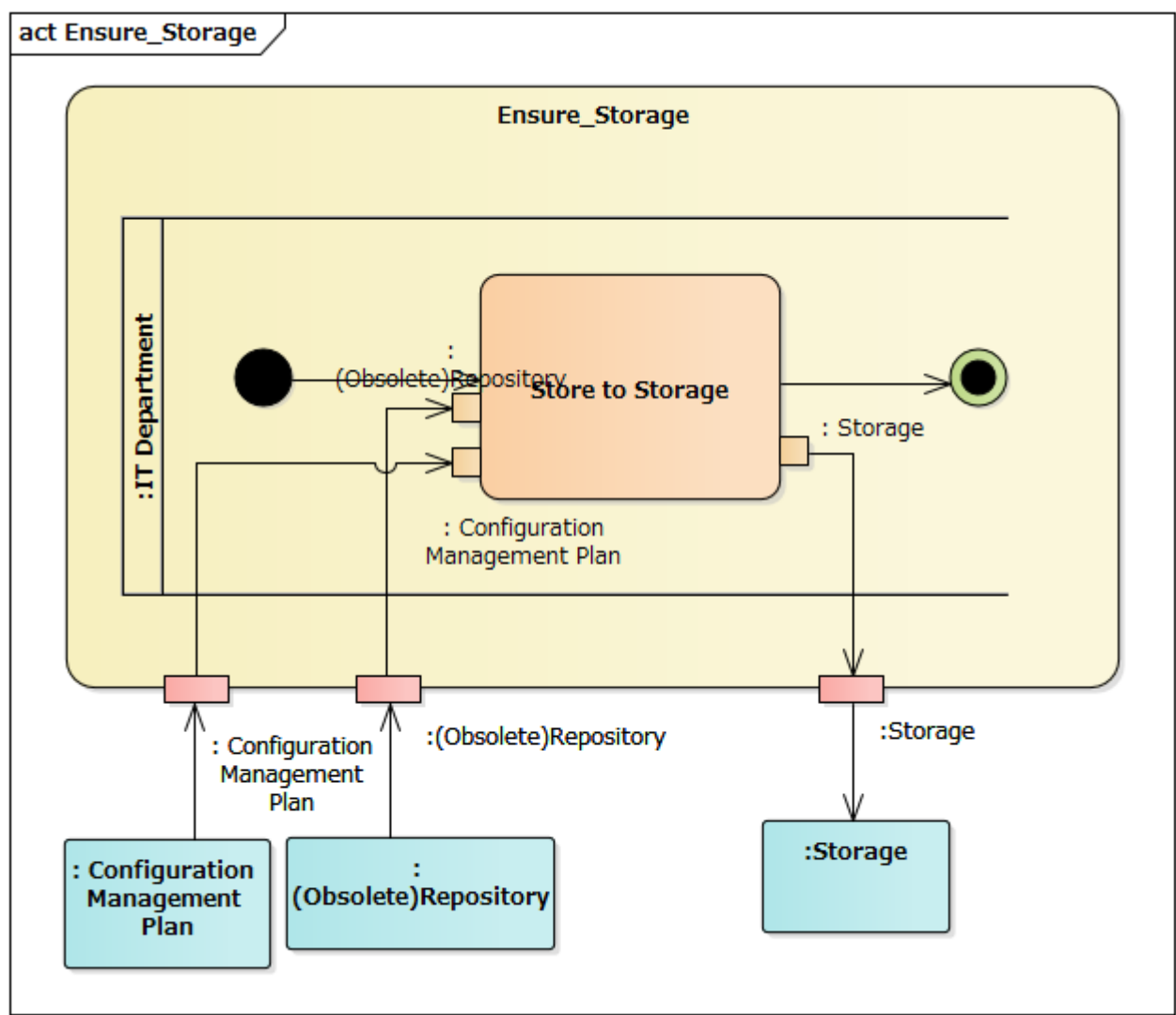
6.4.6 Monitor Access Log



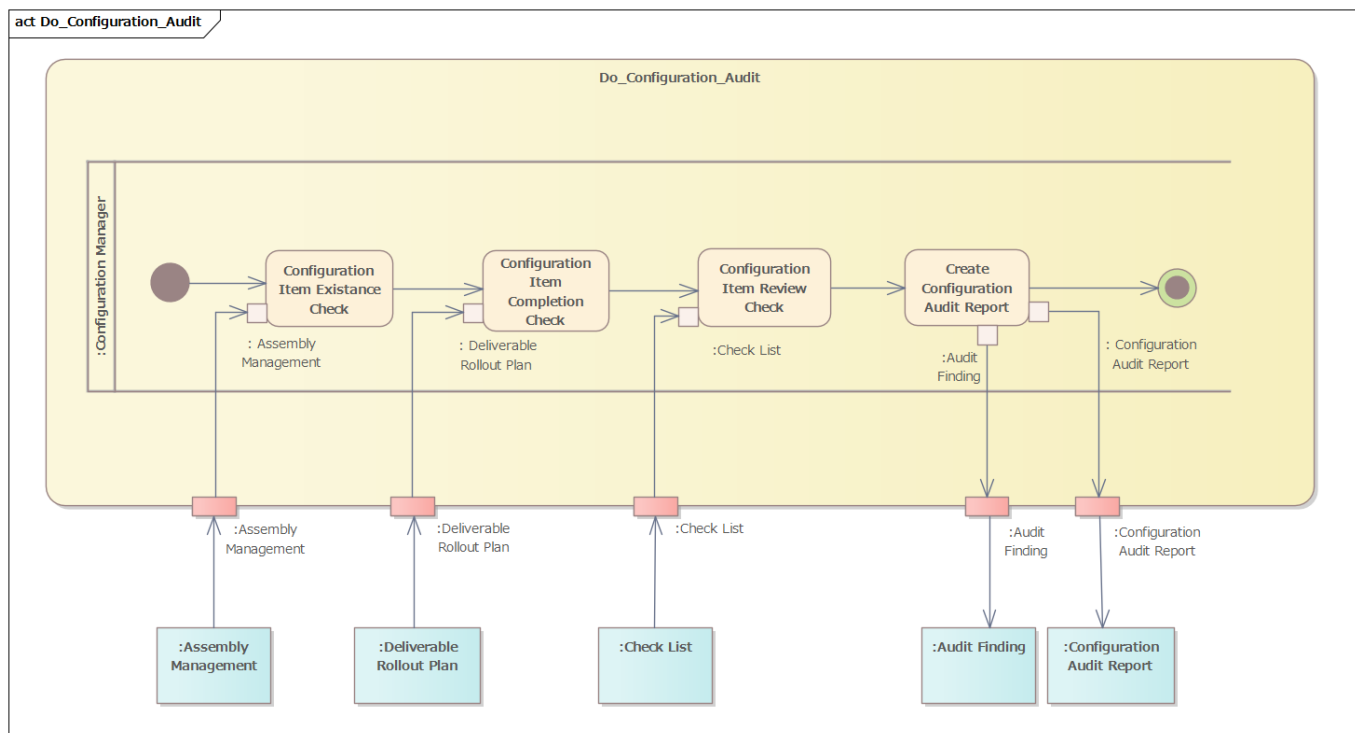
6.4.7 Do Recovery Test



6.4.8 Ensure Storage



6.4.9 Do Configuration Audit

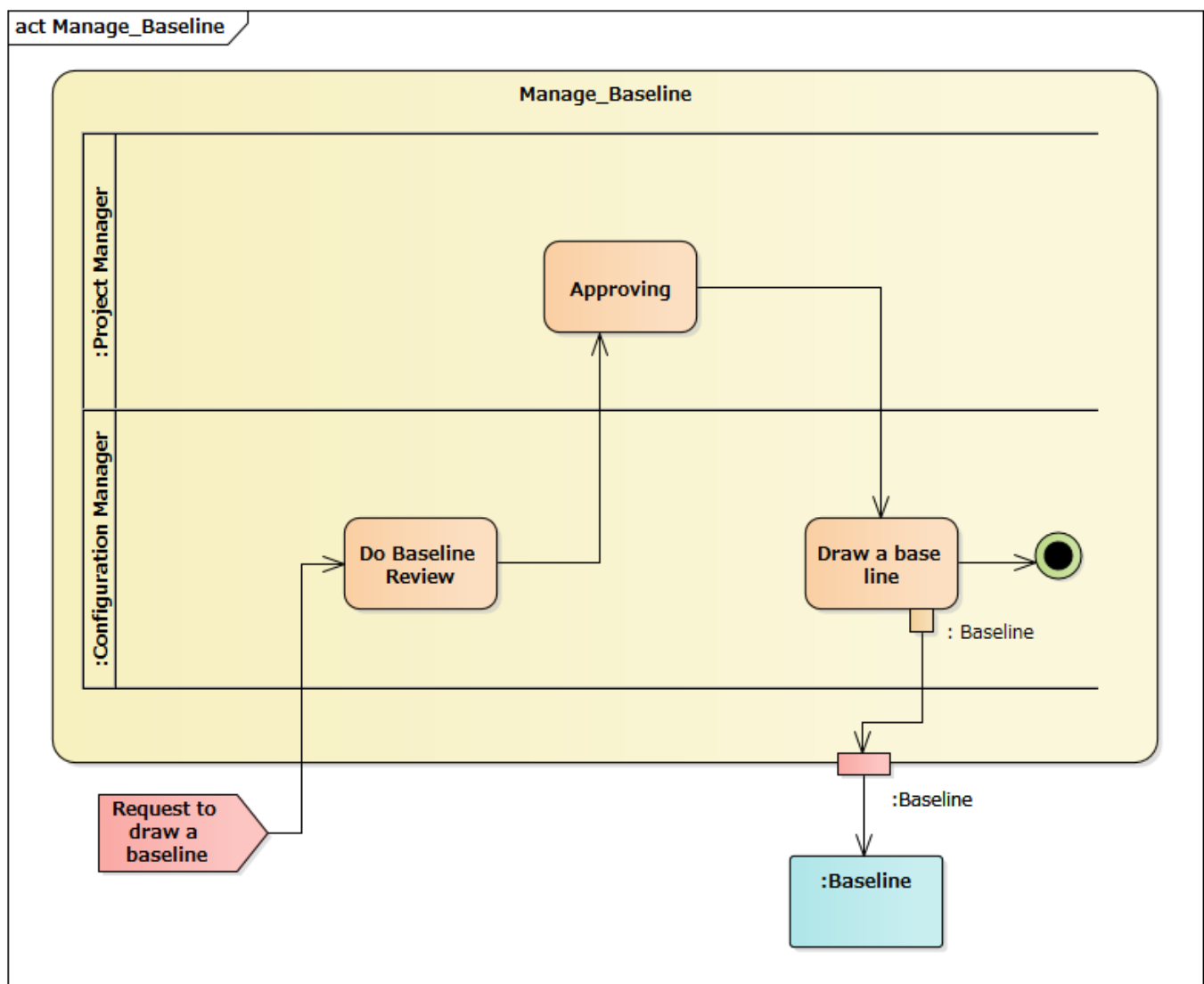


Configuration audit is performed in the following steps:
構成監査は、以下のステップで実施します。

1. Performs a configuration item existence check(The entity of the configuration item exists in the path defined in Assembly Management.)
2. Performs a configuration item completion check(DRP has MG50 or higher.)
3. Checks the review result of the configuration items(Checklist exists. Check result is OK. If there is NG, audit findings ticket has been issued.)
4. Report the check results in the configuration audit report. If there are issues , make audit findings.

1. 構成品目の存在チェックを行います□(assembly managementに定義されたパスに構成品目の実体が存在すること)
2. 構成品目の完了チェックを行います□(DRPのMaturity Gradeが50以上であること)
3. 構成品目のレビュー結果チェックを行います□(Checklistが存在すること□Check結果がOKであること□NGがある場合は、課題ticketが発行されていること。)
4. 上記チェック結果を構成監査レポートで報告します□NGがある場合は、Audit Findingsの発行も行います。

6.4.10 Manage Baseline



7 Infrastructure

To describe here: Define here which Infrastructure is required for carrying out the process.

Tool Name	Description
SVN	It is used to manage the document and source code.
Git	It is used to manage the source code.
EA	
Doors	
3DEX	It is used in Mechanical Domain.
DS2	It is used in Hardware Domain.

8 Performance Measurements

To describe here: Which KPIs are defined for measuring the process performance

Achievement rate of Objective is measured.□

- | | |
|--|--|
| <ul style="list-style-type: none"> • Monitoring <ul style="list-style-type: none"> ◦ Effectiveness KPI□ <ul style="list-style-type: none"> ▪ 1) Achievement Level of SPICE <ul style="list-style-type: none"> • Target Value :□3 ▪ 2) Time to take to restore a development environment. <ul style="list-style-type: none"> • Target Value : Less than a half day • Efficiency KPI□ <ul style="list-style-type: none"> ◦ 1) Time to take to submit a configuration status report. <ul style="list-style-type: none"> ▪ Target Value : Less than 30 minutes ◦ 1) Time to take to create a baseline. <ul style="list-style-type: none"> ▪ Target Value :□Less than 2week | <ul style="list-style-type: none"> • Monitoring <ul style="list-style-type: none"> ◦ プロセスの効果を示す KPI□ <ul style="list-style-type: none"> ▪ 1) A-SPICE Level達成度 <ul style="list-style-type: none"> • Target Value : 3(Level) ▪ 2) 開発環境の復元が速やかにできる <ul style="list-style-type: none"> • Target Value : 半日以内 • プロセスの効率を示す KPI□ <ul style="list-style-type: none"> ◦ 1) 構成ステータスレポートを出力するまでにかかる時間 <ul style="list-style-type: none"> ▪ Target Value :30分以内 ◦ 2) マイルストーンレビュー開催からベースライン登録完了までの期間 <ul style="list-style-type: none"> ▪ Target Value : 2週間以内 |
|--|--|

9 Review

 **To describe here:** Which reviews exists for checking the process performance (e.g. peer reviews) and which Review Criteria are defined for each review.

- | | |
|---|--|
| <ul style="list-style-type: none"> • Baseline Audit <ul style="list-style-type: none"> ◦ Review Criteria • Configuration Audit <ul style="list-style-type: none"> ◦ Review Criteria | <ul style="list-style-type: none"> • ベースライン監査 <ul style="list-style-type: none"> ◦ Review Criteria • 構成監査 <ul style="list-style-type: none"> ◦ Review Criteria |
|---|--|

10 Quality Gates

 **To describe here:** Which formal Q-Gates are relevant for this process

- | | |
|--|---|
| <ul style="list-style-type: none"> • List of Quality Gates <ul style="list-style-type: none"> ◦ QA Audit • Quality Gate passing criteria <ul style="list-style-type: none"> ◦ It is defined in Quality Management Process. • Quality Gate responsible role for that process <ul style="list-style-type: none"> ◦ It is defined in Quality Management Process. | <ul style="list-style-type: none"> • CMの活動は、QAによって実施される品質監査によって評価される。 <ul style="list-style-type: none"> ◦ 活動の詳細は、Quality Management Processに定義する。 |
|--|---|

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